

Data sharing

Dr Cyril Pernet, PhD

As usual ...

- This is required that you engage with the lecture; group exercises for the class are in the

Question boxes

- Of course, stop me at any time if you have questions

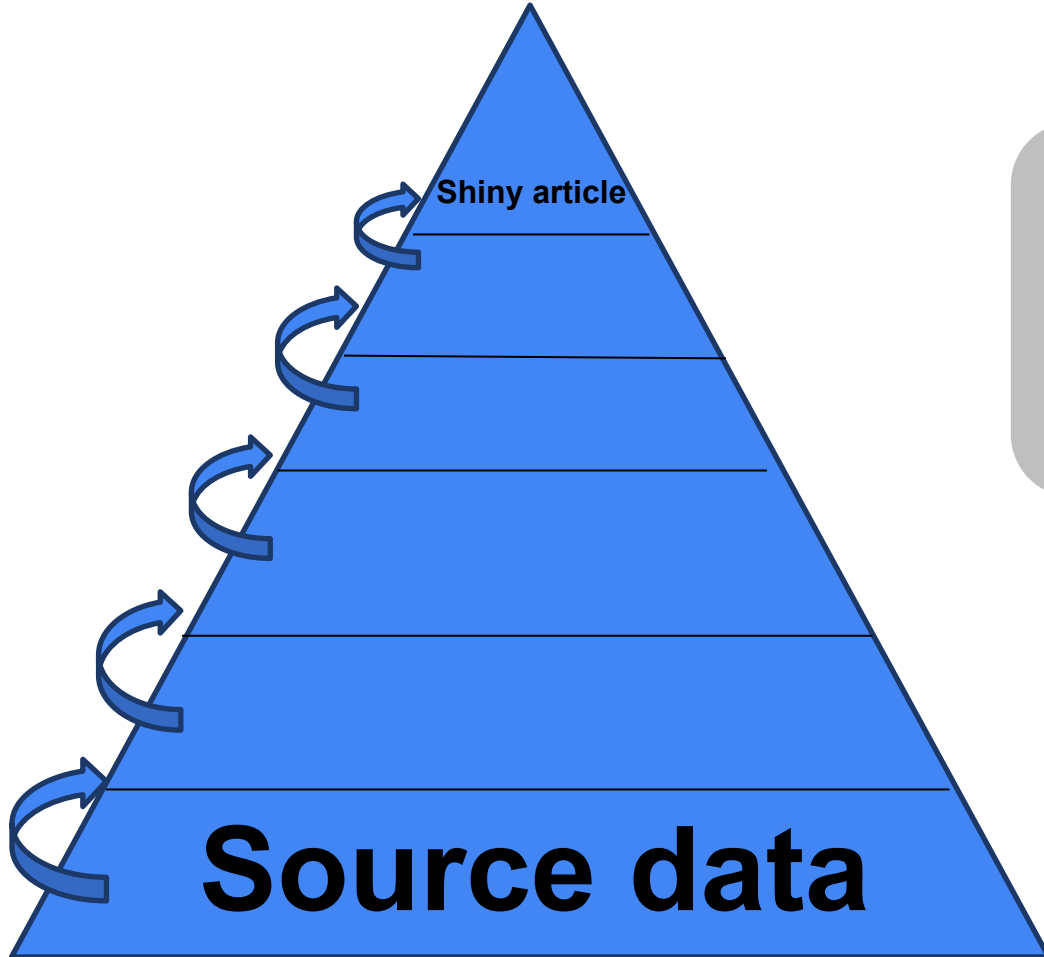
Intended Learning Outcomes

- Recall concepts of good data management making data FAIR and lawful
- Being able to think through and list the different elements of your research that can be shared
- Explain how different form of sharing helps reproducibility and transparency

Scientific Data Management

- What are scientific data?
- For whom is that useful?
- How to share?

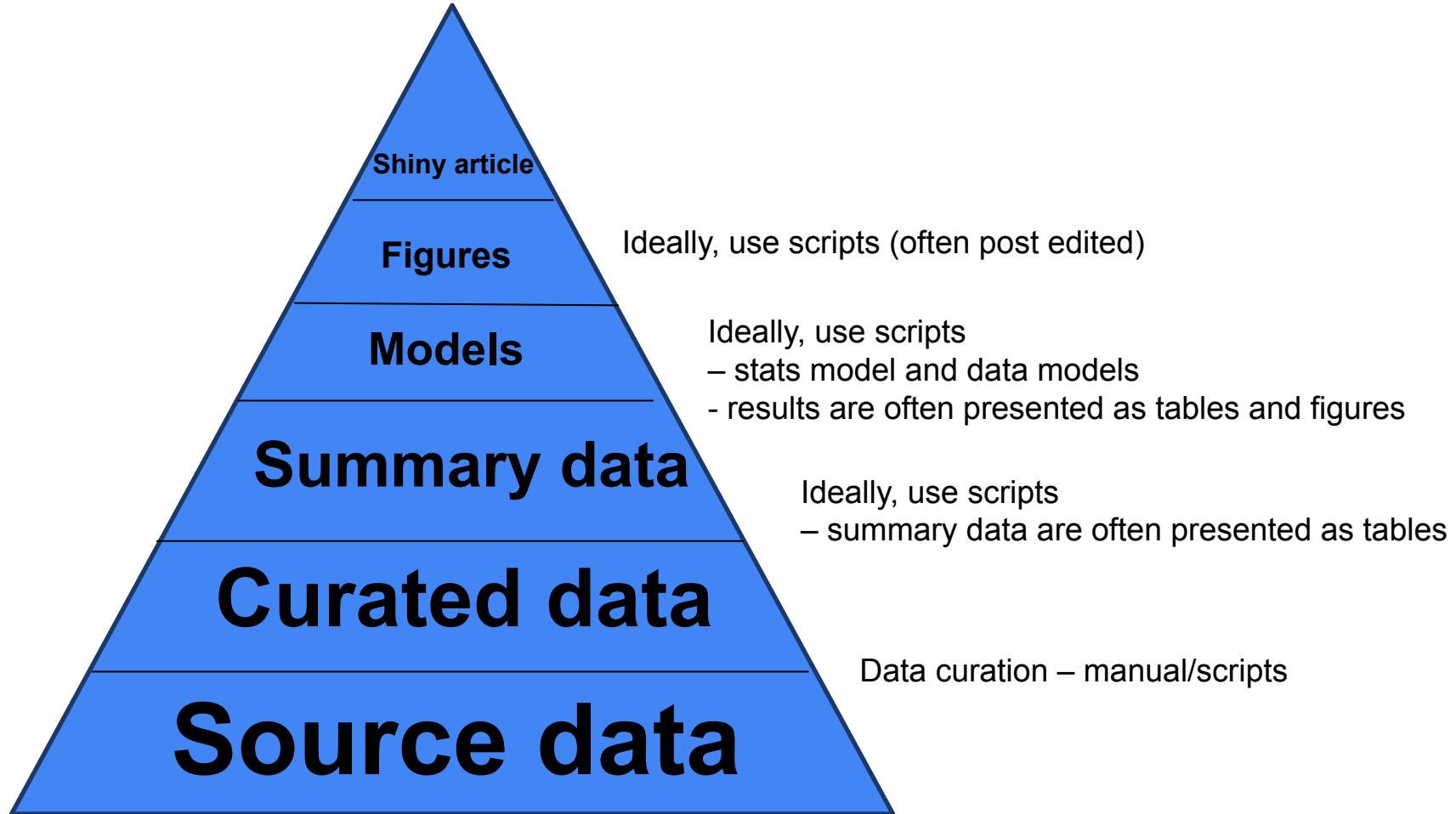




What data?

- Levels/types of data
- how it is made
- usefulness to share

(make groups – let's draw on the board)



Scientific Data Management

- Experimental Data are organized in a consistent way
 - i) for you in 6 months
 - ii) for your collaborators (and PI)
 - iii) for easy data sharing
- Analysis data (code and results) are stored and shared allowing reproducibility using the well-organized experimental data (lecture 1.02)
- Documentation is systematic being in the procedures (lecture 1.03) or the analysis (lectures 2.02, 2.03)
- Open science encompasses a variety of tools and practices that allow good data management and thereby reproducible science

Reminders → FAIR DMP

FAIR Principles

F
A
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The FAIR Guiding Principles for scientific data management and stewardship

Mark D. Wilkinson, Michel Dumontier, IJsbrand Jan Aalbersberg, Gabrielle Appleton, Myles Axton, Arie Baak, Niklas Blomberg, Jan-Willem Boiten, Luiz Bonino da Silva Santos, Philip E. Bourne, Jildau Bouwman, Anthony J. Brookes, Tim Clark, Mercè Crosas, Ingrid Dillo, Olivier Dumon, Scott Edmunds, Chris T. Evelo, Richard Finkers, Alejandra Gonzalez-Beltran, Alasdair J.G. Gray, Paul Groth, Carole Goble, Jeffrey S. Grethe, Jaap Heringa, Peter A.C 't Hoen, Rob Hooft, Tobias Kuhn, Ruben Kok, Joost Kok, Scott J. Lusher, Maryann E. Martone, Albert Mons, Abel L. Packer, Bengt Persson, Philippe Rocca-Serra, Marco Roos, Rene van Schaik, Susanna-Assunta Sansone, Erik Schultes, Thierry Sengstag, Ted Slater, George Strawn, Morris A. Swertz, Mark Thompson, Johan van der Lei, Erik van Mulligen, Jan Velterop, Andra Waagmeester, Peter Wittenburg, Katherine Wolstencroft, Jun Zhao & Barend Mons[✉] - [Show fewer authors](#)

FAIR Principles

Findable
Accessible
Interoperable
Reusable

The FAIR Guiding Principles for scientific data management and stewardship

Mark D. Wilkinson, Michel Dumontier, IJsbrand Jan Aalbersberg, Gabrielle Appleton, Myles Axton, Arie Baak, Niklas Blomberg, Jan-Willem Boiten, Luiz Bonino da Silva Santos, Philip E. Bourne, Jildau Bouwman, Anthony J. Brookes, Tim Clark, Mercè Crosas, Ingrid Dillo, Olivier Dumon, Scott Edmunds, Chris T. Evelo, Richard Finkers, Alejandra Gonzalez-Beltran, Alasdair J.G. Gray, Paul Groth, Carole Goble, Jeffrey S. Grethe, Jaap Heringa, Peter A.C. 't Hoen, Rob Hooft, Tobias Kuhn, Ruben Kok, Joost Kok, Scott J. Lusher, Maryann E. Martone, Albert Mons, Abel L. Packer, Bengt Persson, Philippe Rocca-Serra, Marco Roos, Rene van Schaik, Susanna-Assunta Sansone, Erik Schultes, Thierry Sengstag, Ted Slater, George Strawn, Morris A. Swertz, Mark Thompson, Johan van der Lei, Erik van Mulligen, Jan Velterop, Andra Waagmeester, Peter Wittenburg, Katherine Wolstencroft, Jun Zhao & Barend Mons  - [Show fewer authors](#)

DMP

1) Data

- what data will be collected (type and size)
- what type of analyses will be done
- what derived data will be generated (type and size) vs. need to be archived (might be just summary statistics as csv, new images, a mathematical model)

- 2) How will you document the data? i.e. make metadata describing what this is, how it was collected, analysed and generated - is there a data curation standard?
- 3) What will you share and how?
- 4) How will the data be preserved?

FAIR DMP

Findable and Accessible

How do you achieve that?

FAIR DMP

Findable and Accessible = must be in an indexed repository (your university library? Other repository?)

- Preferably searchable
- Preferably with keywords and other metadata
- Open Access make things easier
- Access controlled is fine too - but by who?
Under which conditions access is granted?

You must have
your data open
access to be FAIR



GDPR Reminder

- When processing Personal Identifiable Information – Art 4(1): data relating to a person leading to direct or indirect identification for instance using physical, physiological, genetic, mental, economic, cultural or social data.
- Animal studies??
 - Dead individuals??

GDPR Reminder

- When processing Personal Identifiable Information – Art 4(1): data relating to a person leading to direct or indirect identification for instance using physical, physiological, genetic, mental, economic, cultural or social data.
- Animal studies, studies from dead individuals, fully anonymous tissue data are all out of scope (no GDPR = no excuse not to share).
- Data from living humans, with no associated metadata (e.g. demographics and health data) and no ID (that is the data have a random ID and you can never go back to figure who this is from) are anonymous.
- Anything else, including pseudonymized data (the *process* of removing identifiers) should be seen as PII.

Infrastructure Requirements

- To be FAIR, sharing PII requires consent to share but also an infrastructure allowing public data sharing (metadata are findable $\neq$ open data) with access control to identified users (accessible) using legal agreement(s) such as a DUA and SCC of non EU users.
- A DMP doesn't have to be FAIR, and many cases you will not have the means to make PII easily findable (need a platform). Using data publication and a dedicated repository you can make data accessible and more easily findable.
- Identify if that is possible early on! Check with your library/institution what is planned, this is their job to provide infrastructure to support research/funders requirement.

FAIR DMP

Interoperability = commonly used and open data formats

Reusable = metadata using international standards and ontologies



Your file format, structure, organisation, naming and documentation (if any) might not be understandable to anyone (lectures 1.02 and 1.03)

Resources

<https://dmponline.dcc.ac.uk/> (there are some country specific versions, google it)

<https://www.openaire.eu/how-to-create-a-data-management-plan>

<https://library.stanford.edu/research/data-management-services/data-management-plans>

<https://howtofair.dk/how-to-fair/metadata/>

Publishing



Conventional

Subscription-based/
Pay wall Protected

Reader-pay model

No free access

© transfer agreement



GREEN

Open Access

Self-archiving after
embargo

Hybrid model

Free access embargoed

© transfer agreement



GOLD

Open Access

Immediate Availability

Author-pay model

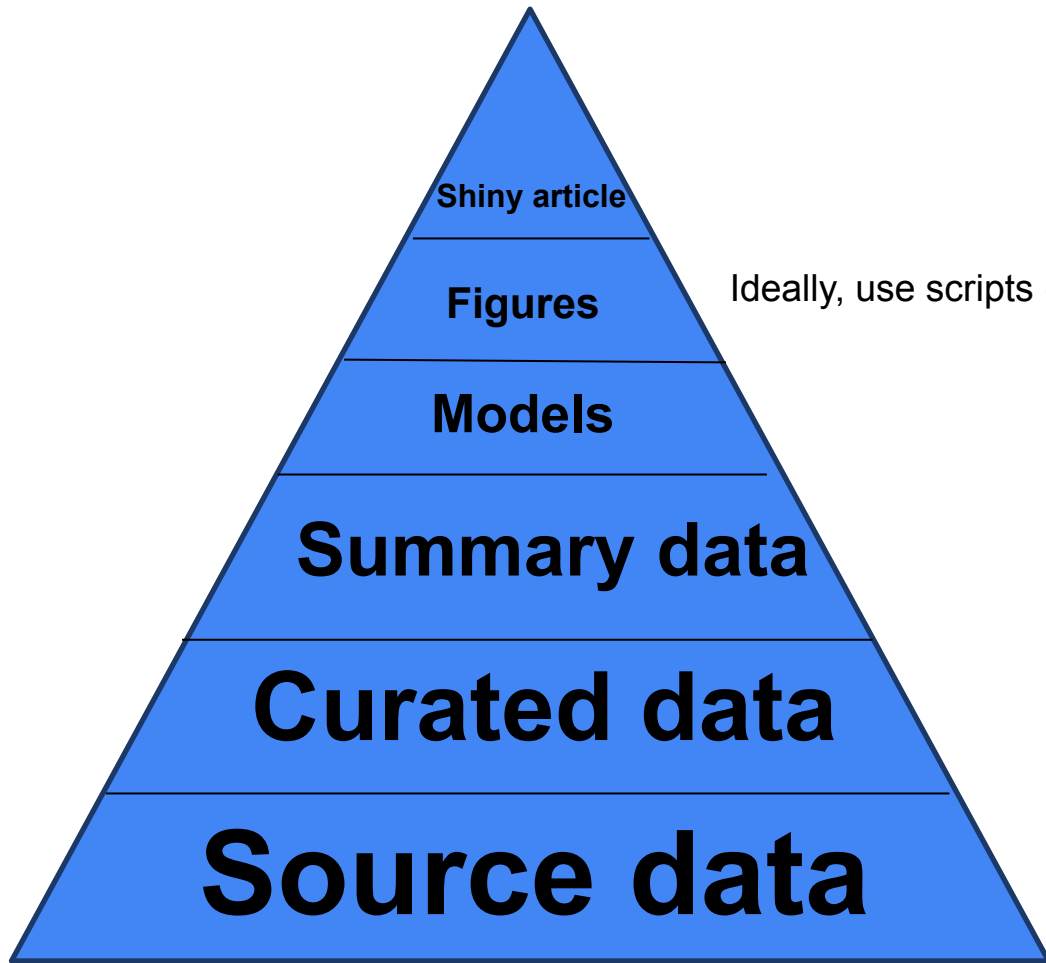
Free access at all times

CC license agreement

+ updated preprint
= gold for way cheaper



By Anke van Eekelen



Ideally, use scripts (often post edited)

Should you use AI for scientific figures?

Xinyu Guo et al., [Cellular functions of spermatogonial stem cells in relation to JAK/STAT signaling pathway](#), *Frontiers in Cell and Developmental Biology* 2024 -- retracted

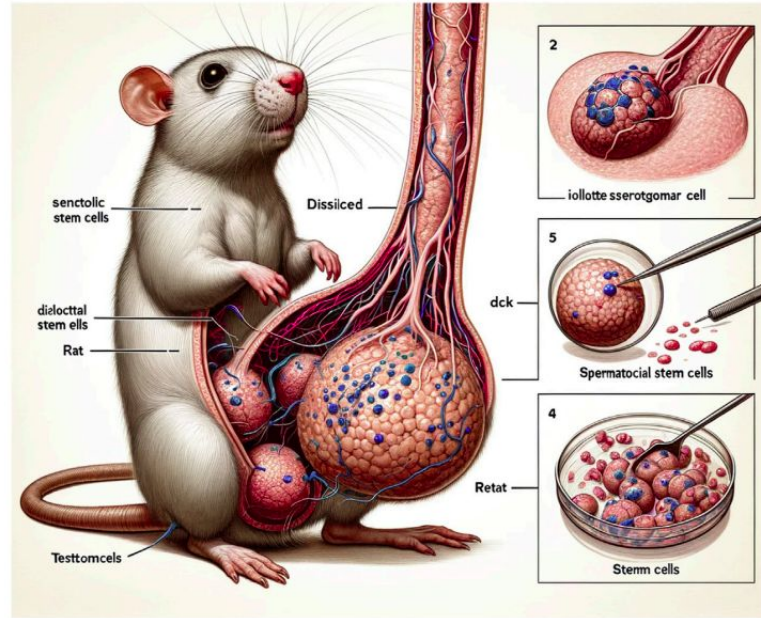


FIGURE 1
Spermatogonial stem cells, isolated, purified and cultured from rat testes.

<https://scienceintegritydigest.com/2024/02/15/the-rat-with-the-big-balls-and-enormous-penis-how-frontiers-published-a-paper-with-botched-ai-generated-images/>

Should you use AI for scientific figures?

Consider use cases: explain vs report

- reporting must come from data (tables, images and code to make the figure) ; generative AI can beautifies it, but sharing derived data and code to make the initial figure ensures transparency
- explaining uses simplified data, fully generated can be used

<https://www.nature.com/articles/d41586-026-02233-w>

Publish and own your figures !! Example:



NeuroImage

Volume 170, 15 April 2018, Pages 348-364



A critical analysis of neuroanatomical software protocols reveals clinically relevant differences in parcellation schemes

Shadia Mikhael^a  , Corné Hoogendoorn^b, Maria Valdes-Hernandez^a, Cyril Pernet^a

Show more 

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<https://doi.org/10.1016/j.neuroimage.2017.02.082> 

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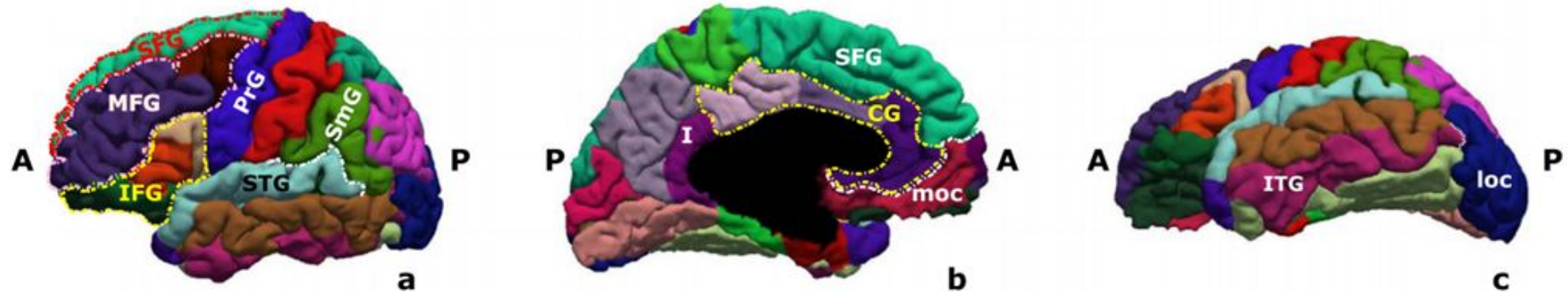
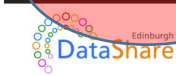


Fig. 4. Lateral (a), medial (b), and inferior (c) cortical surfaces of a 30–35 year-old right-handed male subject, as per FreeSurfer's Desikan-Killiany parcellation protocol. Border precision lacked for the ROIs we investigated, particularly at (1) the PrG's medial border, (2) the SFG and CG's anterior border, (3) the SmG-STG border, (4) the CG-isthmus border, and (5) the ITG-loc border. The raw volume was downloaded from <http://psydata.ovgu.de/studyforrest/structural/sub-01/>. I: isthmus, moc: medial orbitofrontal cortex, loc: lateral occipital cortex (Mikhael, 2017).

Publish your figures before review (and update as needed) - get an open licence and DOI (ideally have csv or related data associated) and include DOIs of your figures in the article

The figures belong to you! An open licence like CC BY says the shared object can be used for profit, so the publisher does not mind using it, but now you can also reuse your figures in other articles, books, for on-line teaching, etc. + get cited because people can use your figures too.



INFORMATION SERVICES

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Examples of Automated Cortical Parcellation

No Thumbnail	Citation Mikhael, Shadia. (2017). Examples of Automated Cortical Parcellation. [image]. University of Edinburgh. Centre for Clinical Brain Sciences. http://dx.doi.org/10.7488/ebs/1963.
Date Available 2017-03-09 Type image	Description The figures presented show examples of anatomical variability and the resulting automated parcellation from different software (also included a diagram of our search for such software). Software packages for cortical parcellation generally combine tools which segment the brain into various regions, or volumes, and provide corresponding measurements, or morphometrics. The results are used for volumetric, region-of-interest (ROI) and connectivity analysis, offering a better understanding of these volumes in the population(s) under investigation. It is therefore crucial to understand how these regions are defined, or the segmentation protocol of a package, as this has significant implications on the analysis that follows. These images reflect the variability in anatomy and in the way software handle this.
Data Creator Mikhael, Shadia Publisher University of Edinburgh. Centre for Clinical Brain Sciences	Relation (is Referenced By) http://dx.doi.org/10.1016/j.neuroimage.2017.03.039 (17.09Kb) Metadata Show full item record

[Download all files](#)

- Figure_3rd (1.70Kb)
- Cortical parcels as seen in the coronal view (97.75Kb)
- Interpretation of the MarsAtlas for 5 regions of interest (ROIs) (1.549Mb)
- Cortical surfaces parcellated by FreeSurfer (3.152Mb)
- Flow diagram identifying the software packages of interest (2.062Mb)
- An example of a double cingulate gyrus and sulcus and its interpretation by various software packages (11.81Mb)
- Examples of sulcal patterns in a middle-aged male (1.234Mb)
- README file (1.763Kb)

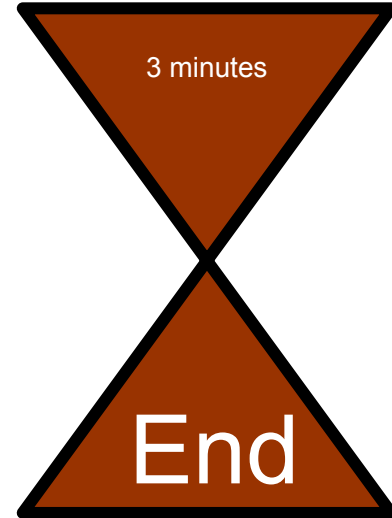
Licensing

What licence?

- Licence versus patent (creator/user agreement vs. prevent usage)
- What do you expect from a licence?

Let see if we can find
something that match your
expectations:

<https://choosealicense.com/>



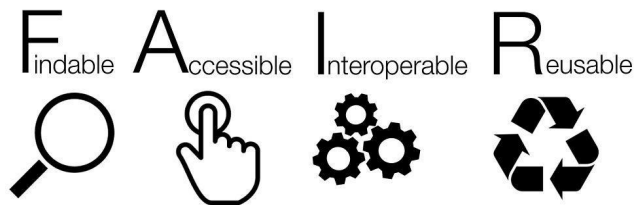
Some Caveats

- Give rights to *use, copy, modify*, and possibly *redistribute*
- MIT (most permissive), Apache, BSD (grant patent to user), GNU GPL (enforce reuse to also be open, not quite like CCNC)
- Your institution will have its own policies and procedures that lay out how to obtain permission to open source your software, you cannot decide – but funders requirements make the difference
- The reuse, bundling or redistribution of code snippets, libraries and other assets such as images as part of your software can only be done if the licences are *compatible*.
- Biomedical data are ‘personal’ = no open licence but Data User Agreement

<https://www.software.ac.uk/resources/guides/adopting-open-source-licence>

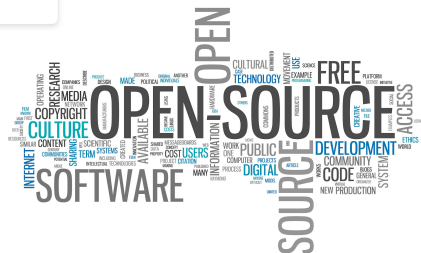
Four aspects to make science open “by design” and not as an after-thought – GigaScience 4

Christmas share



License Types

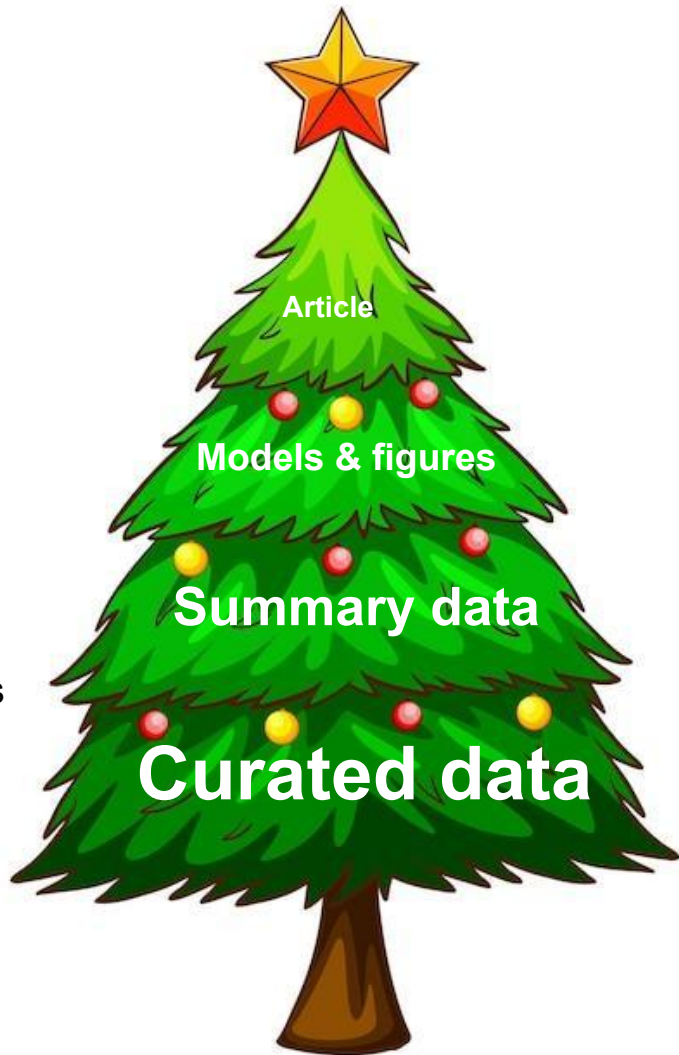
Copyright 'GPL'	permissive MIT, BSD, ...	Creative Commons	Public Domain
viral	attribution	building blocks:	"Don't care"
attribution	protection against liability	- attribution - virality - no commercial use #	liability? ⚡
protection against liability	patents, trademark rules in details	- no derivatives ⚡	



Self-archiving!

Code Sharing
(GitHub)

Data Repositories



Article

Models & figures

Summary data

Curated data

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